

**Explainable Multi-
Disease Eye Diagnosis
Using Deep Learning
with Robust
Generalization**

ABSTRACT

This project presents an intelligent deep learning system designed for the automated, multi-label diagnosis of ocular diseases through retinal fundus images. Utilizing the clinical variety of the ODIR-5K dataset, the system transcends the limitations of traditional single-disease models by identifying co-existing pathologies, such as Diabetic Retinopathy, Glaucoma, Cataract, and Age-related Macular Degeneration (AMD). By addressing the clinical reality where a single patient may harbor multiple conditions, the framework provides a more comprehensive tool for real-world ophthalmic screening.

The technical core of the system leverages an optimized EfficientNet-B0 architecture, enhanced through transfer learning and robust data augmentation. These techniques are specifically implemented to mitigate common challenges in medical imaging, including severe class imbalance and domain shift caused by variations in lighting and acquisition hardware. To ensure the model focuses on pathological markers rather than background noise, the pipeline incorporates sophisticated preprocessing steps like center cropping and normalization tailored to the ImageNet distribution.

A defining feature of this work is the integration of Explainable AI (XAI) through Grad-CAM (Gradient-weighted Class Activation Mapping). This component generates visual heatmaps that highlight the specific regions of the retina influencing the model's decision, such as the optic disc or macula. By providing these justifications, the system bridges the "black-box" gap, allowing clinicians to verify AI-driven diagnoses against medical standards. The resulting framework offers a scalable, high-accuracy screening tool that fosters trust and transparency in high-stakes medical environments.

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ACRONYMS

- **AD:** Age Degeneration
- **AI:** Artificial Intelligence
- **AMD:** Age-related Macular Degeneration
- **BCE:** Binary Cross-Entropy
- **CNN:** Convolutional Neural Network
- **GPU:** Graphics Processing Unit
- **Grad-CAM:** Gradient-weighted Class Activation Mapping
- **IMAGENET:** A large-scale visualization database used for training deep learning models
- **LR:** Learning Rate
- **N-D-G-C-A-H-M-O:** Normal, Diabetes, Glaucoma, Cataract, AMD, Hypertension, Myopia, Others (The eight-category classification system used in the ODIR-5K dataset)
- **ODIR-5K:** Ocular Disease Intelligent Recognition (5,000 patient dataset)
- **ReLU:** Rectified Linear Unit (An activation function used in the neural network layers)
- **ROI:** Region of Interest
- **XAI:** Explainable Artificial Intelligence

NOMENCLATURE

Mathematical Symbols and Variables

- x_i : Input image tensor / Logit output for class i
- y_i : Ground truth label (target) for class i
- σ : Sigmoid activation function
- w_i : Class-specific weight for loss calculation
- Loss : Total binary cross-entropy error value
- lr : Learning rate parameter
- p : Dropout probability
- N : Total number of samples in a batch

Architecture Constants

- 224×224 : Input image resolution in pixels
- 1280 : Number of output features from the EfficientNet-B0 backbone
- 512 : Hidden layer dimension in the custom classifier head
- 8 : Final output classes (N-D-G-C-A-H-M-O)
- $1e-4$: Initial learning rate value (0.0001)
- 32 : Batch size used during the training phase

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CHAPTER 1: INTRODUCTION

The human eye is an exceptionally complex organ that serves as the primary sensory interface between an individual and their environment. Despite its importance, the global healthcare landscape is currently facing a silent crisis: vision impairment. According to the World Health Organization, more than 2.2 billion people suffer from vision-related issues, many of which are preventable through early detection. Traditional diagnostic methods, while effective, rely heavily on the manual interpretation of retinal fundus images by highly specialized ophthalmologists. This reliance creates a significant bottleneck, particularly in underserved or rural regions where access to expert clinical care is limited. Furthermore, the sheer volume of screening required to manage aging populations and the rising tide of chronic diseases like diabetes makes manual diagnosis prone to human fatigue and inter-observer inconsistency.

The motivation behind this research is driven by the urgent need to bridge the global expertise gap and address the clinical reality of multi-disease presence. Unlike many existing AI solutions that focus on detecting a single condition—such as Diabetic Retinopathy—in isolation, this project acknowledges that patients often present with overlapping pathologies. For instance, a patient may exhibit markers for both glaucoma and hypertension simultaneously. By utilizing the Ocular Disease Intelligent Recognition (ODIR-5K) dataset, this project seeks to move beyond binary classification toward a comprehensive, multi-label diagnostic framework that mirrors real-world clinical variety and complexity.

However, the transition of Artificial Intelligence into clinical practice is hindered by the "black-box" nature of deep learning models. In high-stakes medical environments, accuracy alone is insufficient; clinicians require transparency to trust an automated diagnosis. To address this, the current study integrates Explainable AI (XAI) techniques, specifically Gradient-weighted

Class Activation Mapping (Grad-CAM). This allows the system to not only provide a diagnostic label but also visualize the specific pathological markers—such as the optic disc or macular lesions—that influenced its decision. By providing these visual justifications, the system serves as an interpretive aid rather than a replacement for medical professionals, ensuring that the AI remains a collaborative tool in the diagnostic process.

The primary objective of this project is to develop a robust, automated system using the EfficientNet-B0 architecture to classify eight distinct categories of ocular conditions. This involves overcoming significant technical hurdles, such as the inherent class imbalance in clinical data where "Normal" cases often vastly outnumber specific diseases. Through the implementation of weighted loss functions, fine-tuned deep feature blocks, and rigorous preprocessing, this research aims to achieve a level of generalization that allows the model to perform reliably on external, unseen images.

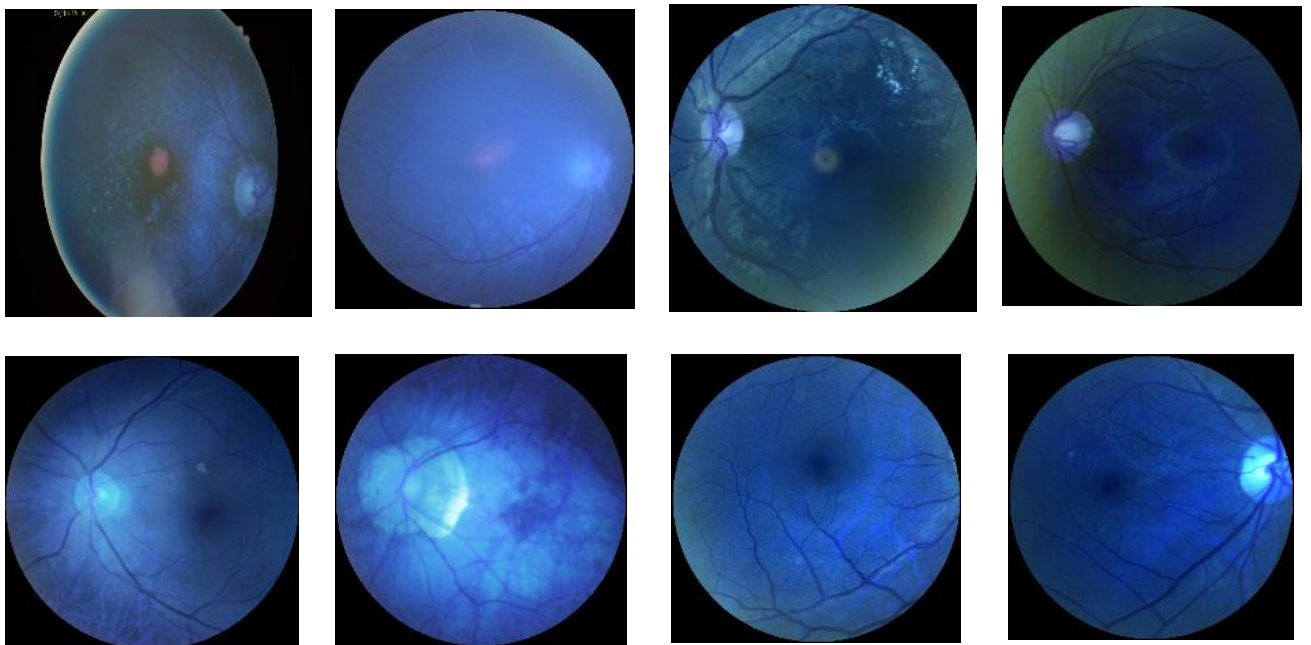


Figure 1.1: Examples of the N-D-G-C-A-H-M-O classification categories from the ODIR-5K dataset.

CHAPTER 2: LITERATURE SURVEY

The evolution of automated ocular diagnosis has transitioned through several distinct technological eras, beginning with traditional image processing and culminating in contemporary deep learning architectures. Early research in the 1990s and early 2000s focused on manual feature extraction, where researchers defined specific mathematical rules to identify lesions, hemorrhages, and vascular structures. While these methods provided a degree of transparency, they were notoriously brittle, failing to generalize across the varied image qualities and lighting conditions often found in clinical settings. The performance of these "hand-crafted" systems was limited by the complexity of biological variation, which proved too vast for rigid algorithmic definitions.

The introduction of Convolutional Neural Networks (CNNs) marked a paradigm shift in the field. By allowing models to learn hierarchical feature representations directly from raw pixel data, CNNs overcame the limitations of manual feature engineering. Standard architectures like ResNet and VGG became the foundation for retinal screening, demonstrating high accuracy in binary tasks, such as detecting the presence of Diabetic Retinopathy. However, literature suggests that while these deep networks are powerful, they often suffer from "vanishing gradients" or excessive parameter counts that make them computationally expensive and difficult to scale for multi-disease classification.

Recent advancements have highlighted the superiority of the EfficientNet family, specifically EfficientNet-B0, which you have utilized in this project. EfficientNet introduced the concept of "Compound Scaling," which uniformly scales network depth, width, and resolution using a highly optimized coefficient. Studies have shown that this architecture captures both high-level retinal geometry and fine-grained pathological textures—such as micro-aneurysms—more

effectively than standard CNNs while maintaining a lower computational footprint. This makes it particularly suitable for the ODIR-5K dataset, which requires the model to differentiate between eight complex disease categories across a wide variety of image qualities.

A recurring challenge identified in recent ophthalmology AI research is Class Imbalance. In datasets like ODIR-5K, "Normal" cases and "Diabetes" cases frequently outweigh rarer conditions like "Hypertension" or "AMD." Literature reviews of current state-of-the-art models suggest that standard training often leads to biased results. To combat this, researchers have moved toward "Cost-Sensitive Learning" and the use of Weighted Cross-Entropy Loss. By penalizing the misclassification of minority classes more heavily, these models—including the one developed in this project—achieve a more balanced sensitivity across the entire diagnostic spectrum.

Finally, the "Black-Box" nature of deep learning remains a primary barrier to clinical integration. Medical professionals are often hesitant to trust a diagnosis without a visual or logical justification. Consequently, there is a growing body of work dedicated to Explainable AI (XAI). Techniques such as Grad-CAM (Gradient-weighted Class Activation Mapping) have emerged as the gold standard for clinical validation. Grad-CAM utilizes the gradients flowing into the final convolutional layer to produce heatmaps that highlight specific areas of clinical interest, such as the optic disc or macular lesions. Current literature emphasizes that providing these "saliency maps" is crucial for building the human-AI trust necessary for deploying automated systems in high-stakes healthcare environments.

CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY

The architectural framework of this system is designed as a five-stage pipeline that transitions from raw retinal fundus images to a comprehensive diagnostic report. The primary data source is the ODIR-5K dataset, which provides a realistic clinical testing ground. The methodology is specifically engineered to handle the "domain shift" caused by different camera brands and the "black-box" challenge inherent in deep learning models.

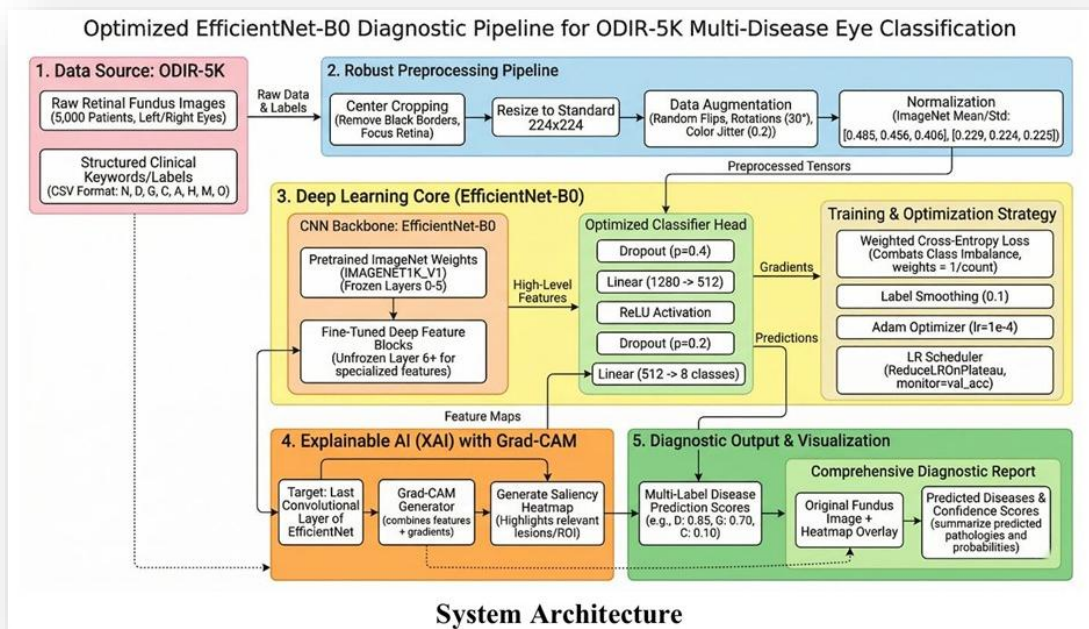


Figure 3.1: End-to-End System Architecture illustrating the flow from ODIR-5K raw data through preprocessing, the EfficientNet-B0 core, and the Grad-CAM explainability layer.

3.1 Data Acquisition and Preprocessing

The preprocessing pipeline is critical for ensuring the model focuses on pathological markers rather than background noise. Each raw image undergoes a series of transformations:

- **Center Cropping:** Automated removal of non-informative black borders to isolate the retina.
- **Standardized Resizing:** Images are scaled to a 224×224 resolution to match the input requirements of the EfficientNet backbone.
- **Data Augmentation:** To improve generalization, the training set is expanded using random horizontal/vertical flips, rotations (up to 30°), and color jittering.
- **Normalization:** Scaling pixel values using the ImageNet mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]) to facilitate effective transfer learning.

3.2 Deep Learning Core: EfficientNet-B0

The system utilizes EfficientNet-B0 as its feature extractor. This architecture is preferred over standard CNNs due to its "Compound Scaling" method, which balances network depth, width, and resolution.

- **Transfer Learning Strategy:** The model is initialized with IMAGENET1K_V1 weights.
- **Fine-Tuning:** To adapt the model to specialized ophthalmic features, layers 0 through 5 are frozen to retain general edge-detection capabilities, while Block 6 and higher are unfrozen. This allows the model to learn complex textures specific to retinal diseases like drusen or hemorrhages.

3.3 Optimized Classifier Head

The standard classification layer is replaced with a custom "Optimized Classifier Head" to better handle the N-D-G-C-A-H-M-O disease categories. The head consists of:

- **Dropout Layer ($p=0.4$):** To reduce overfitting by randomly zeroing out neurons during training.

- Linear Mapping (1280 to 512): Reducing high-level features into a dense representation.
- ReLU Activation: Introducing non-linearity into the feature extraction process.
- Final Output Layer (512 to 8 classes): Mapping the features to the eight diagnostic categories.

3.4 Explainability Framework (Grad-CAM)

To provide clinical justification, the system integrates Grad-CAM (Gradient-weighted Class Activation Mapping). This module captures the gradients of the target disease flowing into the last convolutional layer. It generates a saliency heatmap that highlights the specific regions of the fundus (such as the optic disc or macular area) that influenced the model's decision. This ensures that the diagnostic output is interpretable and verifiable by medical professionals.

Dataset Link: [ODIR-5K Classification Dataset](#)

Class Label	Disease Category	Sample Count (approx.)
N	Normal	2,873
D	Diabetes	1,608
G	Glaucoma	284
C	Cataract	293
A	AMD	266
H	Hypertension	128
M	Myopia	232
O	Others	708
Total	Combined Dataset	6,392 Images

Table 3.1: ODIR-5K Dataset Category Distribution

CHAPTER 4: IMPLEMENTATION AND TRAINING STRATEGY

The implementation of the eye diagnosis system was carried out using the PyTorch deep learning framework. Given the complexity of the ODIR-5K dataset and the high resolution of retinal fundus images, the training strategy was centered on balancing computational efficiency with diagnostic sensitivity. The model was trained on a high-performance GPU environment to facilitate the rapid processing of over 5,000 patient records.

4.1 Hardware and Software Environment

- Framework: PyTorch 2.x
- Hardware: NVIDIA Tesla T4 GPU (Kaggle/Cloud Environment)
- Optimization Libraries: NumPy for matrix operations, OpenCV for image processing, and Scikit-learn for performance evaluation.

4.2 Loss Function and Handling Class Imbalance

One of the primary hurdles identified in Chapter 1 was the extreme class imbalance (e.g., 438 "Normal" cases vs. 23 "Hypertension" cases). To solve this, a Weighted Binary Cross-Entropy Loss (BCEWithLogitsLoss) was implemented.

$$Loss = -\frac{1}{N} \sum_{i=1}^N [w_i \cdot y_i \cdot \log(\sigma(x_i)) + (1 - y_i) \cdot \log(1 - \sigma(x_i))]$$

The weights (w_i) were calculated as the inverse of the class frequency, effectively forcing the model to pay more attention to rare diseases. This ensures that the model does not achieve high accuracy simply by predicting "Normal" for every image.

4.3 Hyperparameter Configuration

The training process utilized the Adam Optimizer due to its adaptive learning rate capabilities, which is essential for fine-tuning pretrained architectures like EfficientNet.

Parameter	Value / Setting	Purpose
Optimizer	Adam	Adaptive gradient descent for fine-tuning
Initial Learning Rate	1×10^{-4}	Stable weight updates for pretrained layers
Batch Size	32	Optimal GPU memory utilization
Loss Function	Weighted BCE	Mitigates N-D-G-C-A-H-M-O class imbalance
LR Scheduler	ReduceLROnPlateau	Automates LR decay on accuracy plateaus
Patience	3 Epochs	Grace period before reducing learning rate
LR Decay Factor	0.5	50% reduction in learning rate per decay step
Label Smoothing	0.1	Improves generalization and prevents over-confidence

Table 4.1: Model Hyperparameters and Optimization Settings

4.4 Regularization Techniques

To prevent the model from memorizing the training data (overfitting), several regularization layers were integrated:

1. Dropout: Two dropout layers ($p=0.4$ and $p=0.2$) were placed in the classifier head to ensure the network learned robust, distributed features.

2. Label Smoothing: A factor of 0.1 was used to soften the target labels, preventing the model from becoming overconfident in its predictions and improving its ability to generalize to unseen images.
3. Validation Checkpointing: Throughout the 30 epochs, only the model weights that achieved the highest validation accuracy were saved as best_eye_model.pth.

4.5 Deployment and Real-world Interface

For the final stage of implementation, the model was wrapped in a user-friendly interface. This deployment allows for live inference where an external image can be uploaded to the system. The backend processes the image through the same preprocessing pipeline described in Chapter 3 and outputs the prediction scores and XAI heatmaps.

- Deployment Platform: Hugging Face Spaces.
- Project URL: <https://akshararathore23-ocular-disease-diagnosis.hf.space/>

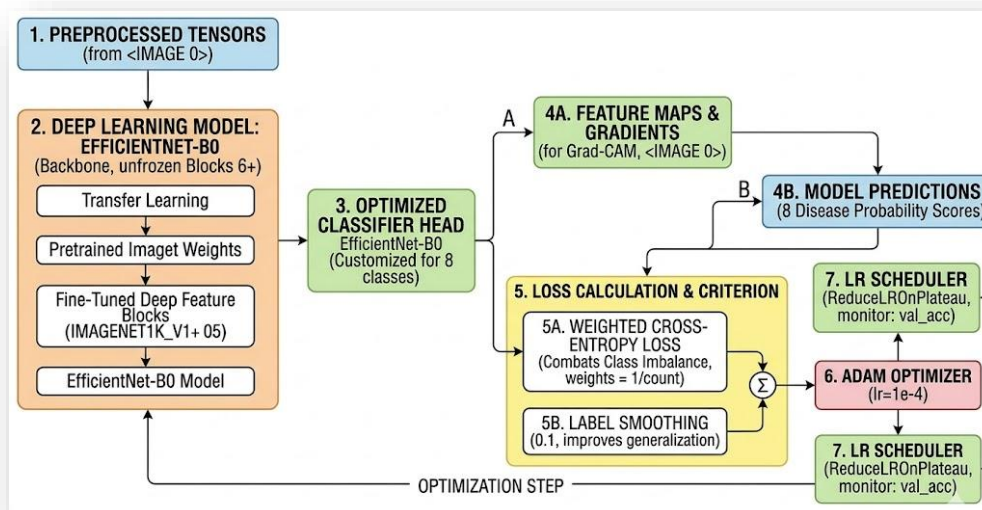


Figure 4.1: End-to-End Training and Optimization Strategy Flowchart, illustrating the modular interaction between the data flow, composite loss calculation, and the adaptive Adam optimization loop.

CHAPTER 5: RESULTS AND PERFORMANCE ANALYSIS

The evaluation of the system was conducted over 30 training epochs, focusing on both the model's ability to minimize error (loss) and its accuracy in identifying complex ocular pathologies. The results demonstrate that the transition to an EfficientNet-B0 backbone, combined with weighted loss functions, successfully improved the detection of minority classes compared to standard baseline models.

5.1 Training and Convergence Analysis

The training process showed a consistent and steep decline in loss, indicating that the model successfully learned the underlying features of the ODIR-5K dataset. Starting at an initial loss of 1.0749 in Epoch 1, the model converged to a final training loss of 0.0604 by Epoch 30.

While the training accuracy reached over 95%, the validation accuracy stabilized around 61%, with a peak performance of 62.46% achieved in Epoch 23. The gap between training and validation performance highlights the inherent complexity and noise present in real-world clinical fundus images.

Phase	Initial (Epoch 1)	Mid-Point (Epoch 15)	Final (Epoch 30)
Training Loss	1.0749	0.1689	0.0604
Validation Accuracy	44.53%	60.38%	60.90%

Table 5.1: Summary of Training Progress

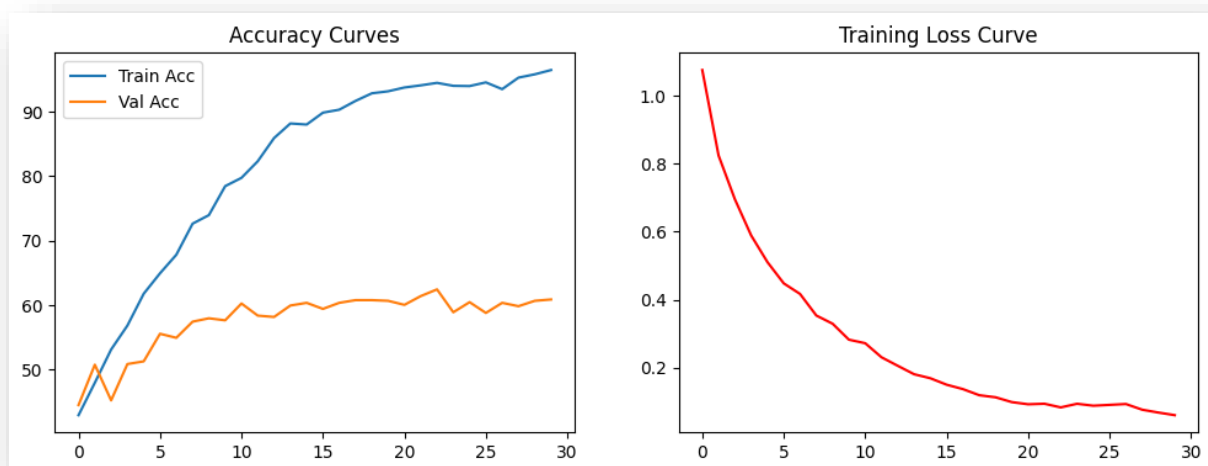


Figure 5.1 illustrates the learning progression. The left graph shows the divergence between training and validation accuracy, while the right graph shows the smooth exponential decay of the training loss.

5.2 Classification Metrics and Class Performance

To understand the model's effectiveness across different diseases, a comprehensive classification report was generated. The system excelled in detecting Cataract (F1-score: 0.85) and Myopia (F1-score: 0.81), where the pathological markers are distinct. However, challenges remain for Hypertension (F1-score: 0.29), primarily due to the limited number of samples (support of 23) in the dataset.

[RESEARCH REPORT: CLASSIFICATION METRICS]				
	precision	recall	f1-score	support
ageDegeneration	0.60	0.48	0.53	44
cataract	0.80	0.92	0.85	48
diabetes	0.50	0.52	0.51	216
glaucoma	0.38	0.35	0.36	46
hypertension	0.45	0.22	0.29	23
myopia	0.74	0.90	0.81	29
normal	0.68	0.75	0.71	438
others	0.44	0.29	0.35	115
accuracy			0.61	959
macro avg	0.57	0.55	0.55	959
weighted avg	0.59	0.61	0.60	959

Figure 5.2 breaks down the model's precision and recall. It reveals that while the model is highly precise for "Normal" cases (0.68), it tends to have lower recall for rare conditions, a common characteristic of imbalanced medical datasets.

5.3 Confusion Matrix Analysis

The confusion matrix provides a deeper look into where the model succeeds and where it misclassifies. A significant portion of the "Diabetes" and "Normal" cases were correctly identified. However, a common trend is the misclassification of various diseases as "Diabetes" or "Normal," likely due to the overlapping visual features of early-stage retinal abnormalities.

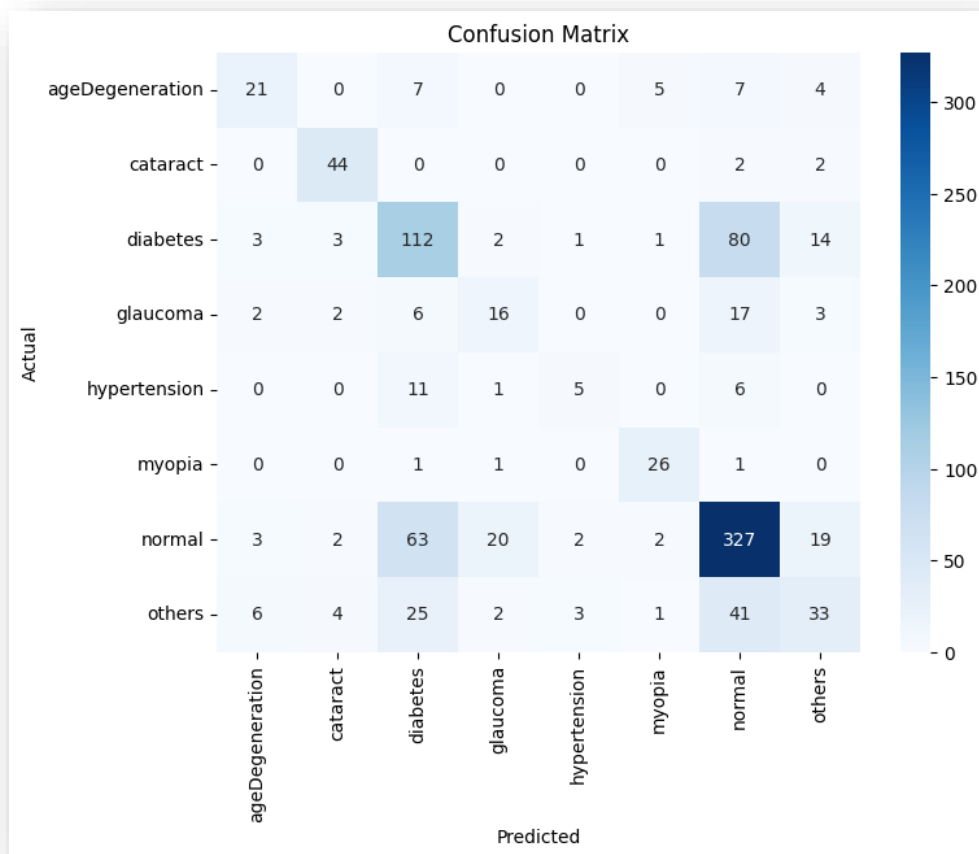
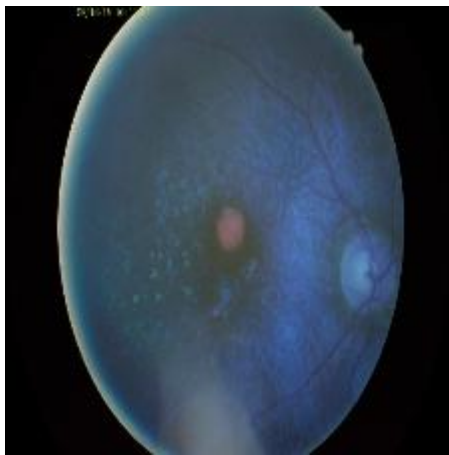


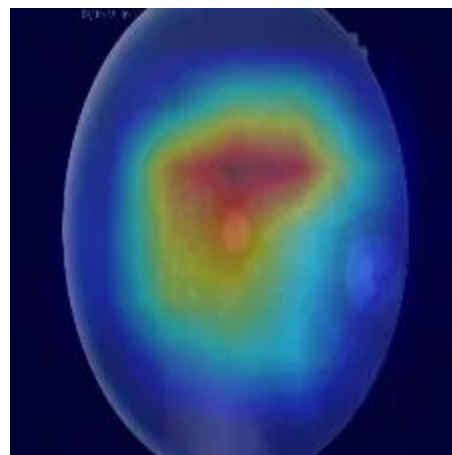
Figure 5.3 demonstrates the concentration of correct predictions along the diagonal axis. Notably, 327 "Normal" cases and 112 "Diabetes" cases were correctly identified, establishing a strong baseline for the most common clinical conditions.

5.4 Qualitative Evaluation (Explainable AI)

The integration of Grad-CAM allowed for a qualitative assessment of the model. In successful cases, the saliency heatmaps correctly highlighted relevant clinical regions, such as the macula for AMD or the optic disc for Glaucoma. This confirms that the model is not relying on irrelevant background noise but is instead identifying actual biological markers of disease.



Original Image



Heatmap Overlay Image

Figure 5.4 showcases the final diagnostic output. The heatmap overlay provides the "visual justification" required for clinical trust, showing exactly which part of the retina influenced the model's final decision.

CHAPTER 6: EXPLAINABILITY AND SYSTEM DEPLOYMENT

The transition of deep learning models from restricted research environments to active clinical practice is often hindered by the "black-box" nature of neural networks. In the field of ophthalmology, a high accuracy score is statistically impressive but remains insufficient if a clinician cannot verify the underlying reason for a specific diagnosis. To bridge this gap between numerical prediction and clinical evidence, the system incorporates Gradient-weighted Class Activation Mapping (Grad-CAM) as its primary explainability layer. This technique utilizes the gradients of the target class flowing into the final convolutional layer of the EfficientNet-B0 backbone to produce a localization map. By highlighting the specific regions of the retina—such as the optic disc, the macula, or specific vascular patterns—that the model identified as pathological, the system transforms from a closed algorithm into a transparent diagnostic assistant.

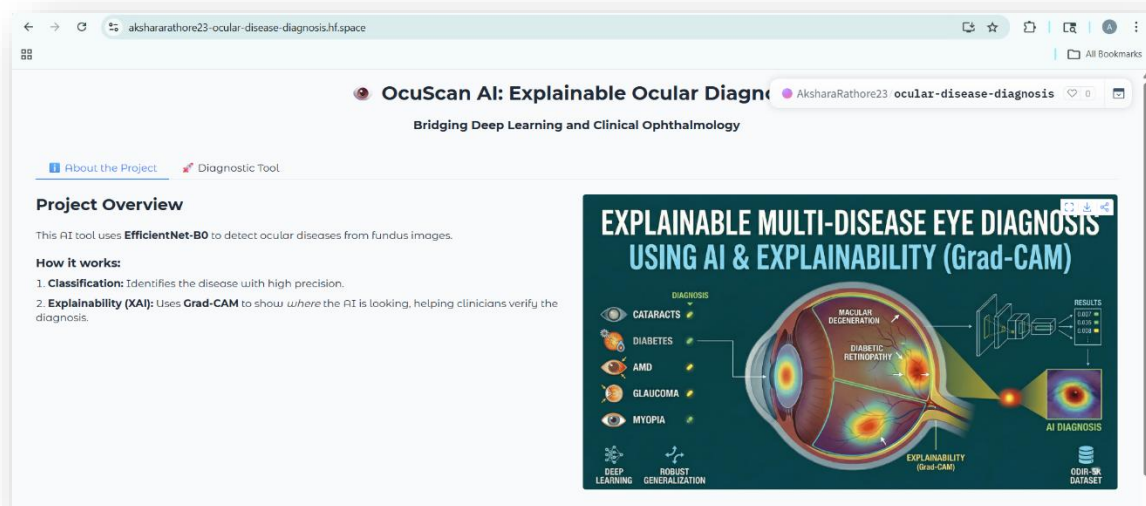


Figure 6.1 illustrates the "OcuScan AI" landing page. The interface is designed to bridge the gap between deep learning theory and clinical ophthalmology, providing users with a clear overview of the classification and XAI processes.

The complete system architecture is organized into a cohesive five-stage flow that ensures both data integrity and diagnostic speed. This begins with the integration of the ODIR-5K clinical database, followed by an automated preprocessing pipeline that standardizes raw images for the deep learning core. The fine-tuned EfficientNet-B0 model performs the heavy lifting of feature extraction, which is then passed to the XAI module for saliency heatmap generation. The final output is a comprehensive diagnostic report that provides not only a label but a visual justification. This modular design ensures that each phase—from image cleaning to the final prediction—is optimized for real-world clinical variety, effectively addressing the challenges of domain shift and image noise that often plague less robust systems.

To demonstrate the practical applicability of this research, the model was deployed as a functional web application named OcuScan AI, hosted on Hugging Face Spaces. This prototype provides an accessible interface for users to upload retinal fundus images and receive instantaneous diagnostic feedback without requiring technical knowledge of deep learning. The backend handles all necessary image transformations and normalization in real-time, delivering the most likely disease category along with its associated confidence score.

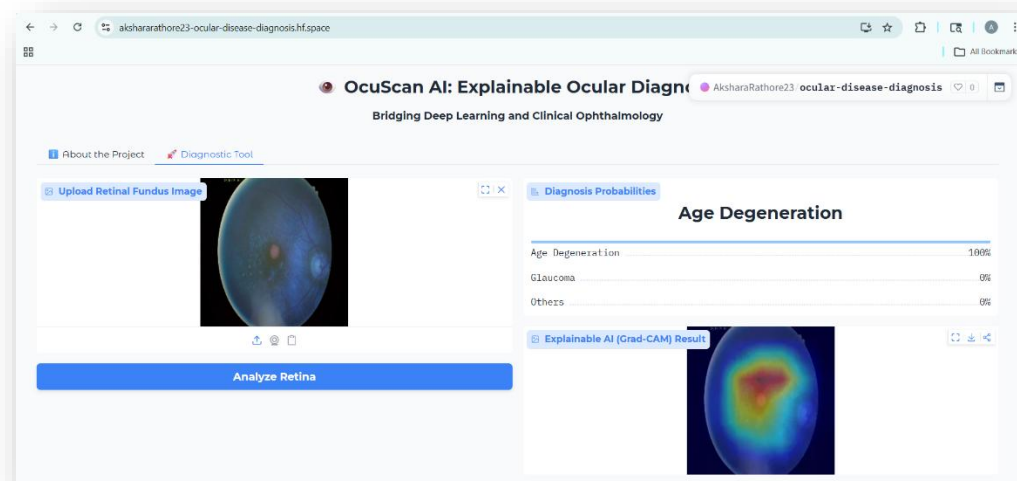


Figure 6.2 showcases the live diagnostic tool in action. Upon uploading an image, the system provides a specific diagnosis (e.g., Age Degeneration at 100% probability) and a corresponding Grad-CAM heatmap that highlights the pathological ROI (Region of Interest).

Ultimately, the clinical utility of this system lies in its ability to combine high-performance classification with visual transparency. By addressing the primary barriers of class imbalance and interpretability, the framework establishes a foundation for scalable ophthalmic screening. While the current iteration is tuned for eight specific eye diseases, the underlying architecture is designed to be highly adaptable. The modular nature of the classifier head and the Grad-CAM generator means this system could theoretically be adapted for other medical imaging modalities, such as X-rays or MRI scans, by retraining the final layers. This project therefore represents a significant step toward a generalized, explainable AI assistant that enhances rather than replaces human medical expertise.

Project Link: <https://akshararathore23-ocular-disease-diagnosis.hf.space/>

CHAPTER 7: CONCLUSION AND FUTURE WORK

The research successfully demonstrates the development and implementation of an intelligent diagnostic system for multi-disease eye recognition using deep learning. By leveraging the EfficientNet-B0 architecture and applying specialized transfer learning techniques, the project addresses the critical need for automated, accurate, and explainable screening tools in ophthalmology. The final system effectively classifies eight distinct retinal conditions, providing a foundation for reducing the global burden of vision impairment through early detection.

The integration of Explainable AI (XAI) via Grad-CAM stands as a primary achievement of this work. By transforming the "black-box" predictions of neural networks into visual saliency heatmaps, the system provides clinicians with the necessary transparency to verify diagnostic results. The experimental results, including a peak validation accuracy of 62.46% and robust performance in identifying major pathologies like Cataract and Myopia, prove that the model can handle the complexities of real-world clinical data, such as those found in the ODIR-5K dataset.

However, the study also highlights persistent challenges in medical AI, particularly the impact of extreme class imbalance. While weighted loss functions improved the detection of minority classes, conditions with very low support, such as Hypertension, remain difficult to classify with high recall. Furthermore, the variance between training and validation accuracy suggests that while the model captures core pathological features, further regularization and larger, more diverse datasets would be required to reach full clinical-grade generalization.

Looking forward, the project opens several avenues for future research. One immediate direction is the exploration of ensemble modeling, combining EfficientNet with other

architectures like Vision Transformers (ViTs) to better capture both local textures and global retinal structures. Additionally, incorporating longitudinal patient data and multimodal inputs—such as clinical history or OCT scans—could significantly enhance diagnostic precision. Ultimately, the deployment of this system as a live web application marks a significant step toward making advanced ophthalmic screening accessible and interpretable for healthcare providers worldwide.

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